

The day the fleet lost its horizon

A first-person lifecycle record of fourteen paper-bot runs, the uncertain AMD submit, the account recovery, and the Internet / IB Gateway disconnect that ended the session.

OFFLINE - CURRENT BROKER TRUTH UNKNOWN

Last broker-verified state at 14:25:32 CT: PAPER / CLEAN / FLAT / 0 positions / 0 working orders. All fourteen runs were retired before the outage.

14	8	25	2
bot runs	max concurrent	strategy round trips	recovery exits

Decision: no third wave. The AMD replacement ticket remained an unsubmitted draft. Reconnect reconciliation is required before any new order-capable action.

Prepared from durable local evidence on 2026-07-27 | Times shown in Central Time

The safe boundary

The experiment stopped from a safe last-known state, not from an active fleet. The important caveat is temporal: the final broker receipt was current when written, but it became historical when Internet and Gateway connectivity were lost.

Last authoritative broker receipt

Receipt **acct-recon-DUM284968-1785180332714-fea6adc105274768**, generated 14:25:32 CT, proved PAPER account DUM284968, state CLEAN, flat exposure, no positions, no working orders, and a passing account reconciliation gate.

Surface	Last durable state	Offline meaning
Fleet	14 run bindings RETIRED	No experiment bot should be order-capable.
Account	Last verified PAPER / CLEAN / FLAT	Safe last-known state, not current truth.
Orders	Last broker snapshot: 0 working orders	Must be re-queried after reconnect.
Journal	Net {}, residual {}, verdict clean	Durable local ownership is balanced.
Clerk	Generation 80 / accepting	Local phase only; broker reachability unknown.
Browser	AMD replacement deploy form	Draft only; Deploy & run was not clicked.

What the panel should have said

OFFLINE - BROKER TRUTH UNKNOWN. Last verified 14:25:32 CT: PAPER / CLEAN / FLAT / 0 working orders. All starts and submits are locked. Reconnect reconciliation is required.

The Clerk's local accepting phase must never be presented as proof of broker connectivity.

A compressed timeline

CT	Moment	Evidence-backed outcome
11:23	Preflight	Clean worktree; paper-only safety boundary established.
11:30	Control restored	Host daemon and Account Clerk recovered from a flat/no-order boundary.
11:40-12:10	Wave one	Six independent bots launched; UI lifecycle defects repaired as found.
12:12	Ownership defect	New bots claimed sibling positions in local portfolios; rollout stopped.
12:25-12:37	Withdrawal	Six runs ended, reconciled flat, and were retired for source repair.
12:48-13:09	Wave two	Eight repaired bots launched independently; ownership isolation held.
13:13	Unknown submit	AMD caller timed out; durable intent was acknowledged and filled after process death.
13:14-13:15	Fail closed	Seven sibling bots halted on account truth; no blind retry occurred.
13:29	Exact recovery	META share closed through its owning namespace.
13:41	Emergency recovery	Confirmed paper account flatten closed the late AMD share.
13:42	Ledger repair	12,529 event rows resequenced; cross-runtime lock defect diagnosed.
14:12-14:19	Fleet retired	Every experiment run reached RETIRED.
14:25-14:26	Safe close	Broker proved flat; Clerk ownership projection became clean.
~14:27	Connectivity loss	Internet and IB Gateway unavailable; third wave not launched.

The operator tells the story

- 1 I began with a paper-only boundary. The host daemon was down, so I proved the broker account flat and free of working orders before restoring the host and Clerk.
- 2 The first wave found product defects in hydration, Resume, polling, reconciliation actions, and incident copy. More importantly, six concurrent bots revealed that a new process could copy sibling account-net exposure into its own portfolio. I stopped rollout rather than exercise a dangerous flatten.
- 3 I retired the first wave from a flat boundary, repaired startup ownership to use the Clerk's per-instance projection, and launched eight repaired bots one at a time. All eight reached On-duty proof.
- 4 AMD then submitted behind the serialized Clerk queue. The caller timed out one second before the broker acknowledgement. I kept the intent outcome unknown and did not retry. The late fill was recovered by exact identity; account safety halted the siblings.
- 5 I closed META by exact namespace recovery and AMD by the typed paper-account emergency workflow. Recovery then exposed cross-environment Clerk routing, account-ledger locking, zero-holding display, and sick-bay retirement defects. Each received a focused regression fix.
- 6 By 14:19 every run was retired. The broker still proved clean and flat at 14:25; the journal was clean at 14:26. I opened an AMD replacement draft but did not submit it. When connectivity disappeared, I stopped instead of treating cached green state as current truth.

The right safety decision

A timeout was not translated to failure. A stale broker snapshot was not translated to current truth. A draft deployment was not translated to a launched bot.

Wave one: six first lives

SPY - first life

11:40:34 launch | run a6901134... | 4 round trips | retired

"I traded one SPY share through four complete paths: 738.42/738.25, 736.52/736.31, 736.62/735.96, and 736.71/736.26. The host ended me with code -9 at 12:25:27, so my ending was recorded EXITED_UNVERIFIED. I left no net SPY position."

QQQ - first life

11:43:27 launch | run 997b036e... | Stop/Resume exercised | 3 round trips

"I was stopped flat during diagnosis. Direct Start correctly refused my STOPPED latch, but the panel had no Resume control. After that UI was repaired, I resumed and completed 677.88/677.78, 678.40/677.17, and 678.01/677.65. I left no net QQQ position."

AAPL - first life

12:06:23 launch | run 9d5fd4a4... | 2 round trips | retired

"I completed two AAPL paths, 336.33/335.86 and 335.90/336.24. I shared the 12:25:27 host ending, was recorded unverified, and left no AAPL position."

MSFT - first life

12:08:04 launch | run 88fc0404... | 3 round trips | ownership witness

"I completed 391.33/391.59, 392.20/391.69, and 392.15/391.70. My local portfolio then claimed a sibling SPY share. That was not a broker trade, but it was dangerous ownership. I became evidence for the fleet withdrawal and left MSFT flat."

NVDA - first life

12:09:35 launch | run b625faac... | 1 round trip | ownership witness

"I bought at 195.71 and sold at 195.58. My sidecar falsely claimed sibling SPY and QQQ exposure while the Clerk attributed those fills correctly elsewhere. I left NVDA flat."

AMD - first life

12:10:50 launch | run cda86645... | 2 round trips | ownership witness

"I completed 479.76/480.50 and 480.19/479.53. I also inherited a sibling QQQ claim, proving the defect grew as bots joined. I ended at 12:25:27 and left AMD flat."

Wave-one result

15 complete strategy round trips. Broker order ownership was correct, but bot-local position ownership was unsafe. Startup was changed from account-net positions to Clerk-attributed per-instance exposure.

Wave two: repaired lives, part I

SPY - second life

12:48:15 launch | run 457f899f... | 2 round trips | safety-halted

"I did not inherit sibling exposure. I completed 737.81/737.60 and 737.79/737.86. When AMD became uncertain, account truth stopped further submission. I halted at 13:15:05 and left SPY flat."

QQQ - second life

12:51:18 launch | run 8f6d1d80... | 2 round trips | next entry blocked

"I completed 678.94/679.31 and 680.01/679.72. My next ENTER was refused because retired-owner live exposure made account truth unsafe. I left QQQ flat."

AAPL - second life

12:53:50 launch | run f8108132... | 2 round trips | next entry blocked

"I completed 335.37/335.59 and 335.74/335.74. Account truth refused my next ENTER after the AMD uncertainty. I left AAPL flat."

MSFT - second life

12:55:45 launch | run dbb68f9a... | 2 round trips | safety-halted

"I completed 393.42/393.75 and 394.11/393.69. The account gate then stopped me rather than allow uncertain sibling exposure to coexist with new risk. I left MSFT flat."

Wave two: repaired lives, part II

NVDA - second life

12:57:51 launch | run 9ee7c054... | 1 round trip | next entry blocked

"I completed 196.92/196.93. My next ENTER was blocked by the account-wide truth gate. I halted at 13:14:05 and left NVDA flat."

AMD - second life

12:59:52 launch | run 2175071b... | intent iuBkB5axRbiUzi41uhaCbw | recovered

"At 13:13 I was told to buy one AMD share near 483.45. My request waited in the serialized Clerk queue. My caller timed out at 13:13:36; the broker acknowledged one second later and filled at 483.41 after I was dead. Nobody retried me. The paper emergency workflow sold the share at 481.99. I ended flat, but I proved the fleet submit deadline was too short."

AMZN - only life

13:04:58 launch | run 16ab4d27... | 1 round trip | next entry blocked

"I bought and sold one AMZN share at 232.09. My next ENTER was refused after AMD became uncertain. I halted safely and left AMZN flat."

META - only life

13:09:00 launch | run eb64b4ae... | entry plus exact recovery | retired

"I bought one META share at 595.27 just before the account-wide halt. I could not complete a strategy exit. At 13:29:01, exact namespace recovery sold me at 594.79 using execution 00025b45.6a6e66d0.01.01. I ended flat."

Fleet trading total

52 strategy fills = 25 complete strategy round trips plus unmatched META and AMD entries. Two identity-preserving recovery sells restored the paper account to flat.

What was tried

Attempt	Result	Meaning
Host recovery after flat proof	Worked	Daemon and Clerk restored; broker reverified.
14 independent UI launches	Worked	No hidden cohort or batch mutation.
Fresh saved-state default	Failed -> fixed	Blank ticket could require a missing sidecar.
QQQ Stop / Resume	Partial -> fixed	Stop worked; required Resume was absent.
Two-second fleet polling	Failed -> fixed	Healthy host sometimes rendered resting.
Six-bot ownership	Unsafe -> fixed	Bot portfolios imported sibling exposure.
Repaired eight-bot ownership	Worked	No replacement imported sibling exposure.
15-minute steady state	Incomplete	All eight held about 4 minutes 34 seconds.
30-second fleet submit	Failed -> fixed	Queue wait consumed AMD caller deadline.
Blind retry of AMD	Avoided	Unknown identity was preserved; no duplicate order.
Exact META recovery	Worked	Owning namespace closed the share.
Paper emergency flatten	Worked	Late AMD share closed; broker proved flat.
Cross-runtime ledger lock	Failed -> fixed	Host and VM created duplicate sequences.
Event sequence repair	Worked	12,529 rows repaired; backup retained.
Closed holding display	Failed -> fixed	Zero quantity and sentinel looked open.
Sick-bay retirement	Failed -> fixed	Disabled generic action swallowed retirement.
Automatic sick-bay roll call	Failed -> fixed	Panel tried to revive terminal bots.
Third-wave revalidation	Not attempted	Connectivity loss made stopping correct.

What the disconnect revealed

1. Green is not a state

Design finding

Every account claim needs observed_at, a freshness deadline, and a visible stale transition. After TTL, the label must become Last known flat - not Flat.

2. Connection is a vector

Design finding

Browser, frontend, backend, host, Clerk, Gateway, IBKR, market-data, account, and order callback links can fail independently. One Connected badge cannot summarize them.

3. Unknown outcomes outlive processes

Design finding

A timeout remains outcome unknown until order_ref, perm_id, and exec_id reconciliation proves accepted, rejected, cancelled, or absent beyond a defined broker horizon.

4. Desired state must survive offline

Design finding

A safe-stop request should become a durable QUEUED - NOT YET APPLIED command. The UI must not claim the bot stopped until an authoritative receipt says so.

5. History needs multiple clocks

Design finding

Event time, broker time, arrival time, and journal receipt time need distinct labels. One callback arrived out of order, and the AMD emergency execution time field is five hours ahead of receipt chronology.

Elegant target state

A durable control plane with an explicit **CURRENT / STALE / UNKNOWN / RECONNECTING** truth model, a connection epoch, fail-closed mutation admission, identity-preserving unknown intents, and mandatory reconnect reconciliation.

Robustness plan

P0 - before the next market-hours fleet

- **Truth-state machine:** model every critical source as CURRENT, STALE, UNKNOWN, or RECONNECTING and attach a connection epoch.
- **Offline interlock:** lock Deploy, Start, Resume, and new strategy submits when account or Gateway truth is stale.
- **Last-known banner:** show disconnect time, last broker observation, positions/orders snapshot, and stale age.
- **Reconnect gate:** reconcile account identity, PAPER proof, positions, orders, executions, Clerk projection, registry, desired states, and unresolved intents before admission.
- **Fresh validation:** re-run the queue budget, cross-runtime mutex, closed-holding, sick-bay retire, and roll-call fixes in a clean paper fleet.

P1 - make recovery operable

- Generate each bot's first-person life story directly from durable events.
- Merge account events using separately labeled event, arrival, and receipt times.
- Persist offline safe-stop requests and show queued versus applied state.
- Add a read-only Fleet Safe Boundary page: bots, unknown intents, residual ownership, positions, working orders, and source freshness.
- Export a signed incident bundle from the control panel.

P2 - prove resilience continuously

- Run deterministic loss drills for browser, backend, host, Gateway, market-data, account-callback, and ack/fill paths.
- Mark critical disconnects within 5 seconds and never admit a new intent after the last healthy epoch.
- Retain unresolved submits through process restart and reach terminal reconnect classification within 60 seconds when the broker responds normally.

The next disconnect drill

- 1 Start eight one-share paper bots and prove one clean connection epoch.
- 2 Create one deliberately delayed but idempotent paper submit in a test harness.
- 3 Remove network access for 90 seconds.
- 4 Within 5 seconds, every surface shows OFFLINE - TRUTH UNKNOWN; Deploy, Start, Resume, and new submits are locked.
- 5 The delayed intent remains OUTCOME UNKNOWN with the same identity; no duplicate order is created.
- 6 A safe-stop request during the outage is durable and visibly unapplied.
- 7 Restore network access; create a new epoch and reconcile identity, positions, orders, executions, Clerk projection, and desired states.
- 8 If the delayed order filled, attribute it and present exact owning recovery. If not, prove absence without blind retry.
- 9 Unlock starts and submits only after a Clean receipt.
- 10 Export a bundle whose bot stories and account timeline agree with durable journals.

Release gate

The system is not robust because it reconnects. It is robust only when it can prove what happened during the gap and refuse new risk until that proof is complete.

Evidence anchors

Last broker reconciliation	acct-recon-DUM284968-1785180332714-fea6adc105274768
AMD unknown intent	iuBkB5axRbiUzi41uhaCbw
META recovery execution	00025b45.6a6e66d0.01.01
AMD emergency execution	0000dc8f.6b293439.01.01
Emergency operation	db07a090-0b97-4bda-be2e-6e2078701fd0
Journal cures	Clerk sequences 1840 and 1841

Full evidence narrative: docs/audits/deployment-validation-connectivity-incident-2026-07-27.md

Research and education only. Nothing in this report is financial advice.